

Module 2

Measurement of angles & tapers

Introduction

- The angle is defined as the opening between two lines which meet at a point.
- If one of the two lines is moved around a point in an arc, a complete circle can be formed.
- The circle is divided into 360 equal parts and each part is called as degree($^{\circ}$).
- Each degree is divided in 60 minutes ($'$) and each minute into 60 seconds($''$).

Objectives

After studying this unit, you should be able to

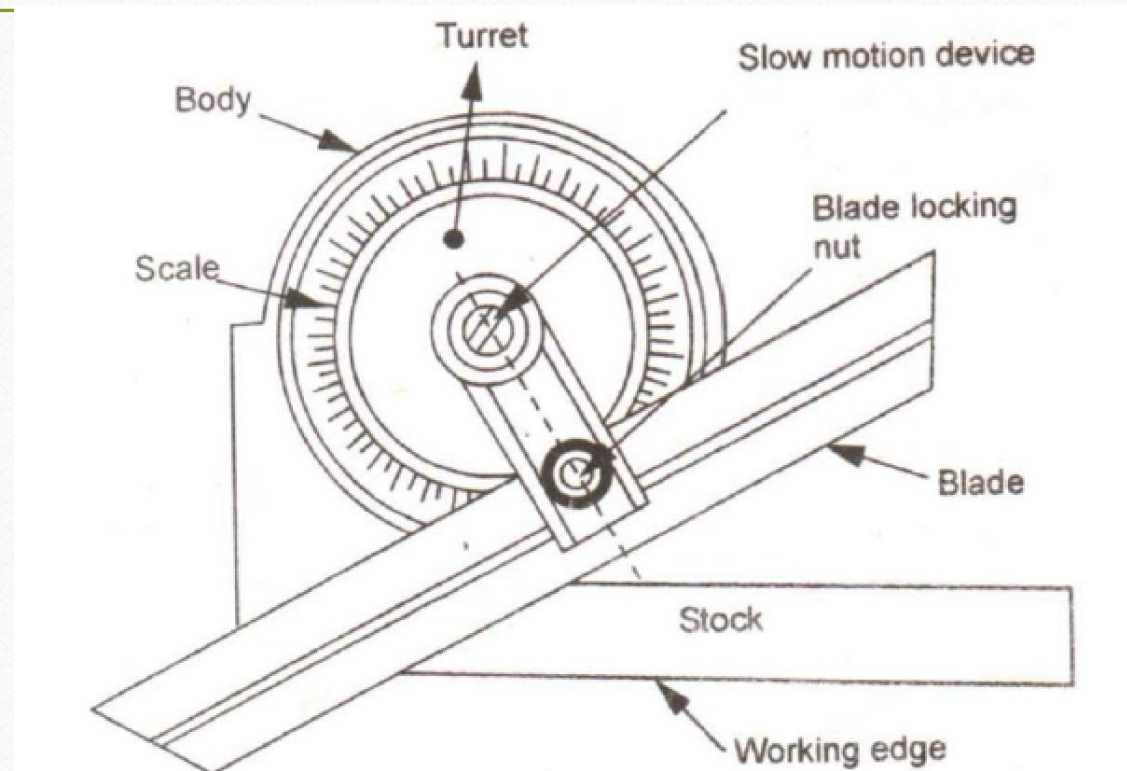
- Familiarize yourself with various types of angular measuring devices, and
- Choose a suitable angular measuring device according to the precision required

LINE STANDARD ANGULAR MEASURING DEVICES

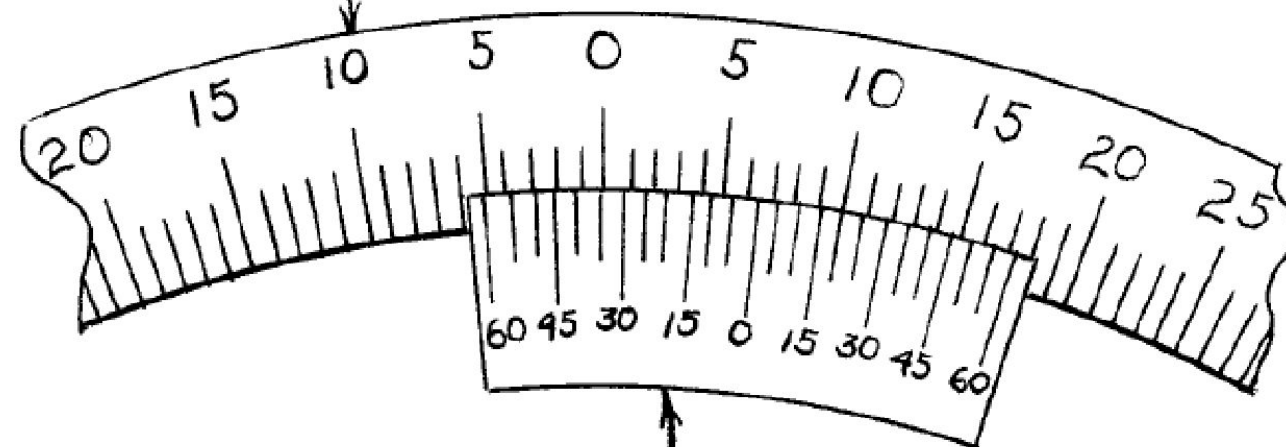
- **Vernier Bevel Protractor (Universal Bevel Protractor):**
 - instrument for measuring the angle between two faces of a component.
 - capable of measuring angles to within 5 min.

COMPONENTS- BEVEL PROTRACTOR

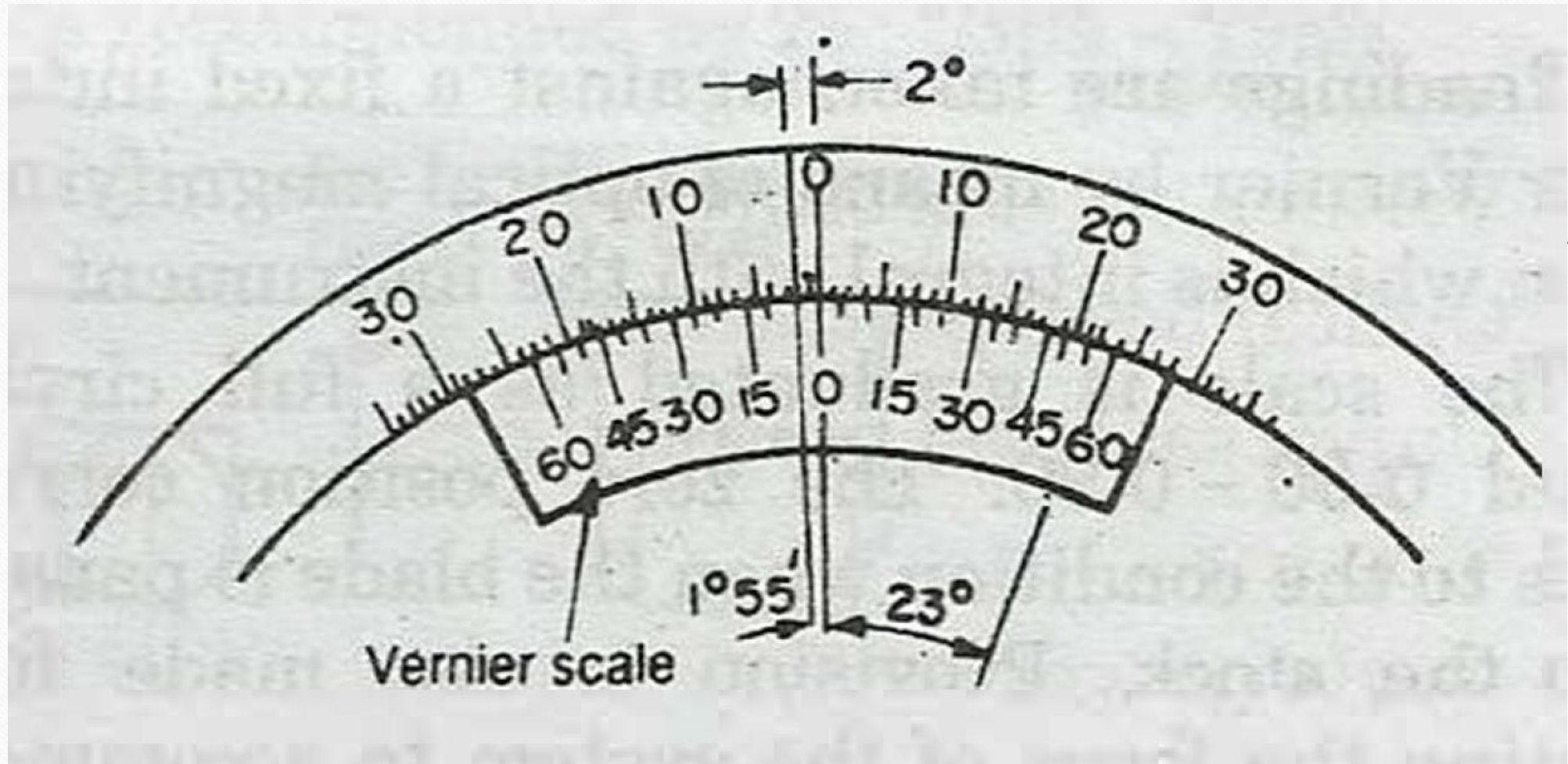
- BODY
- STOCK
- BLADE



DEGREE SCALE



VERNIER SCALE



- As shown in Fig. the main scale is graduated in degrees of arc. The Vernier scale has 12 divisions each side of the centre zero. These are marked 0-60 minutes of arc, so that each division equals $\frac{1}{12}$ of 60, that is 5 minutes of arc. These 12 divisions occupy the same space as 23 degrees on the main scale. Therefore, each division of the Vernier is equal to $\frac{1}{12}$ of 23° or $1\frac{11}{12}^\circ$.

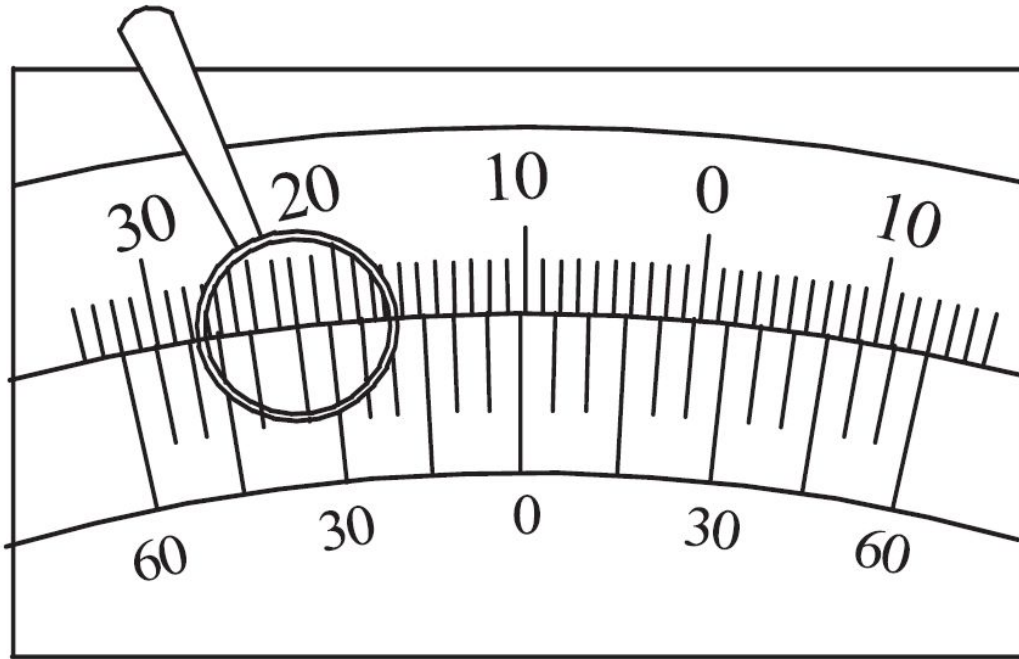
Since two divisions on the main scale equals 2 degrees of arc, the difference between two divisions on the main scale equals 2 degrees of arc, the difference between two divisions on the main scale and one division on the Vernier scale is $2^\circ - 1\frac{11}{12}^\circ = \frac{1}{12}^\circ$ or 5 minutes of arc.

Let us assume that the zeroth division on both the main and the vernier scales are lined up to coincide with each other. Now, as the dial rotates, a vernier division, starting from the fifth minute up to the 60th minute, progressively coincides with a main scale division until the zeroth division on the vernier scale moves over the main scale by 2° .

Therefore, the least count is the difference between one vernier division and two main scale divisions, which is $1/12^\circ$ or $5'$.

Reading Vernier Scales

Reading Vernier Scales

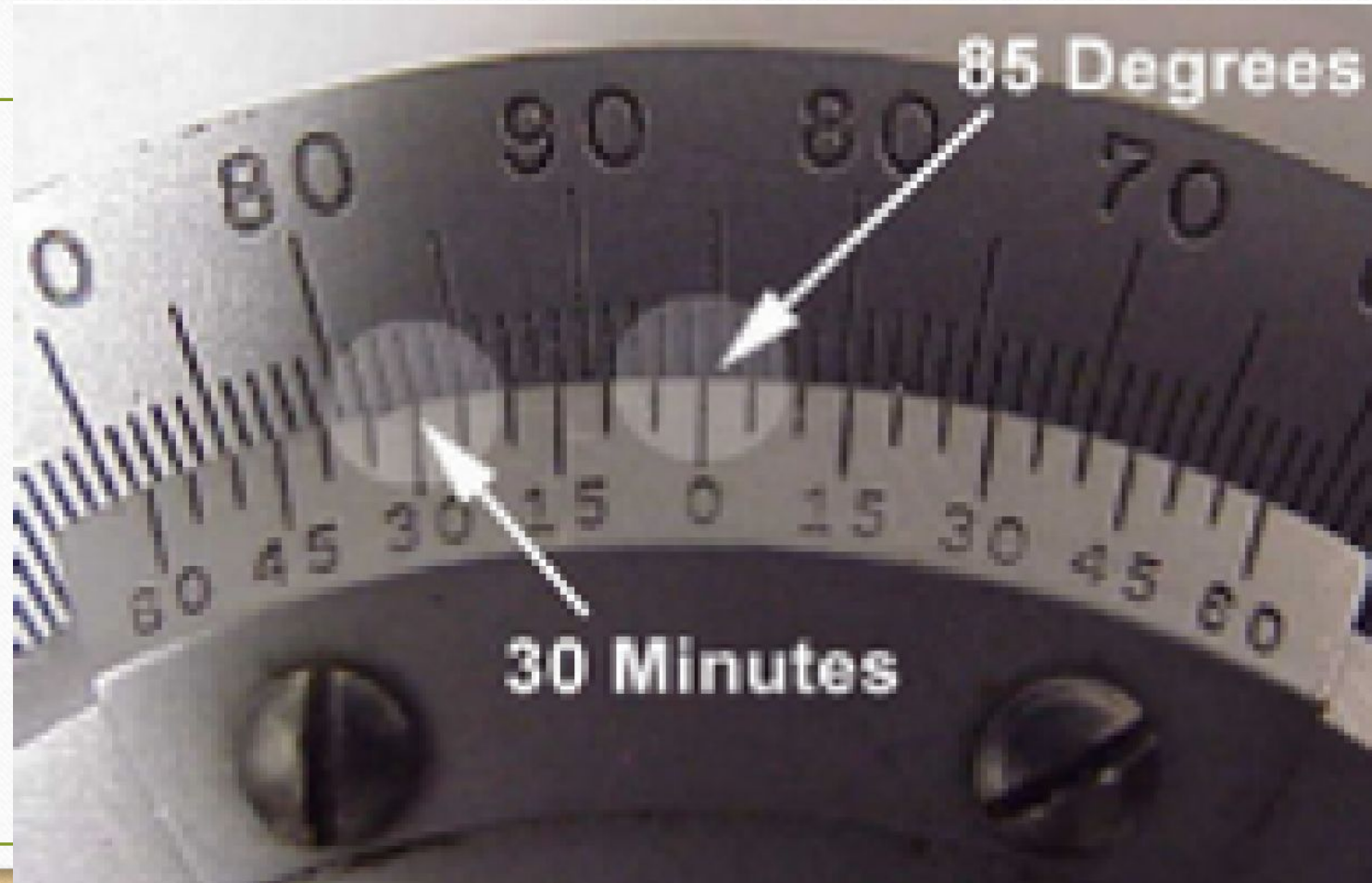


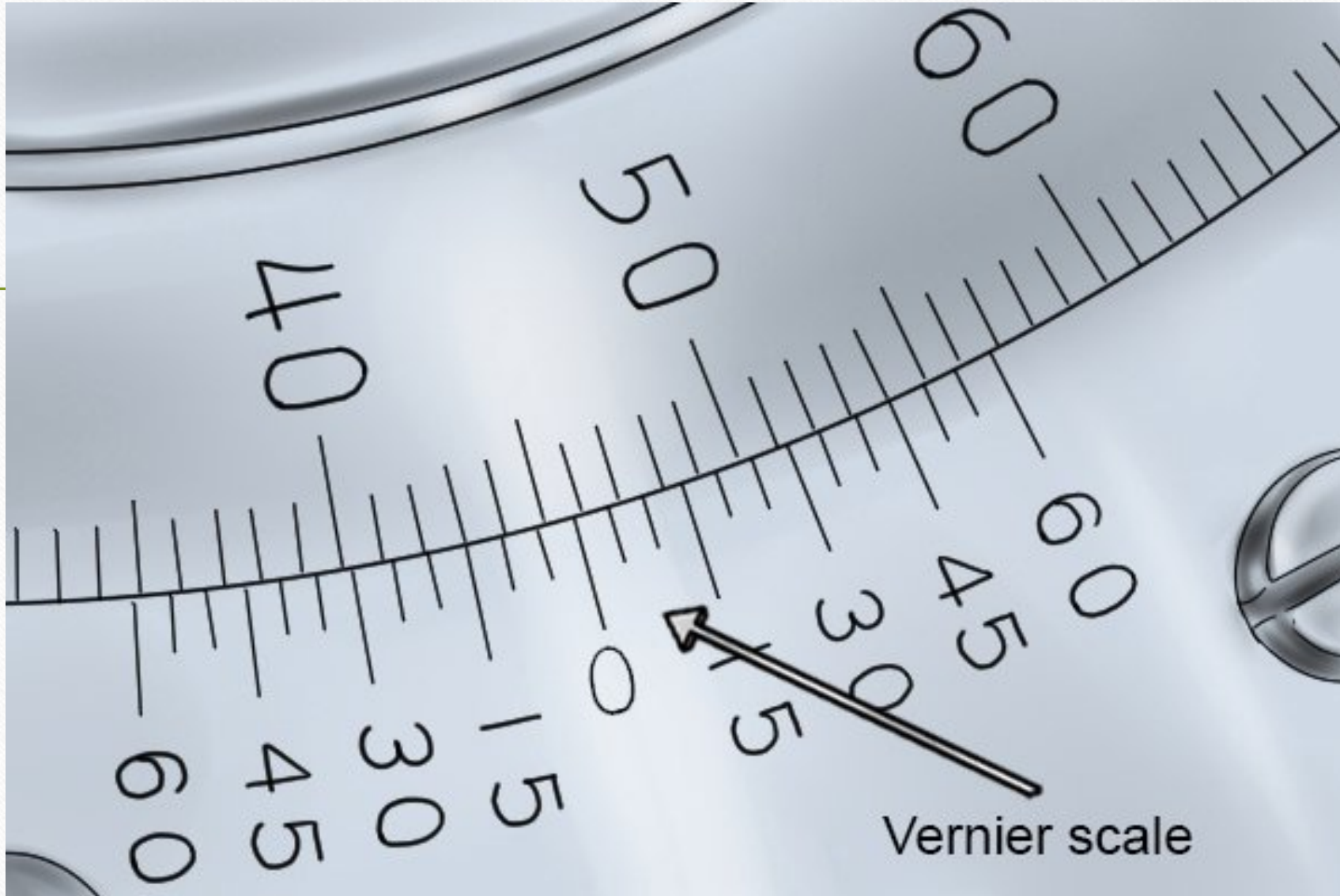
The zeroth division of the vernier scale is just past the 10° division on the main scale. The seventh division, marked as the $35'$ division, on the left hand side of the vernier scale coincides with a division on the main scale. Therefore, the reading in this case is $10^\circ 35'$.

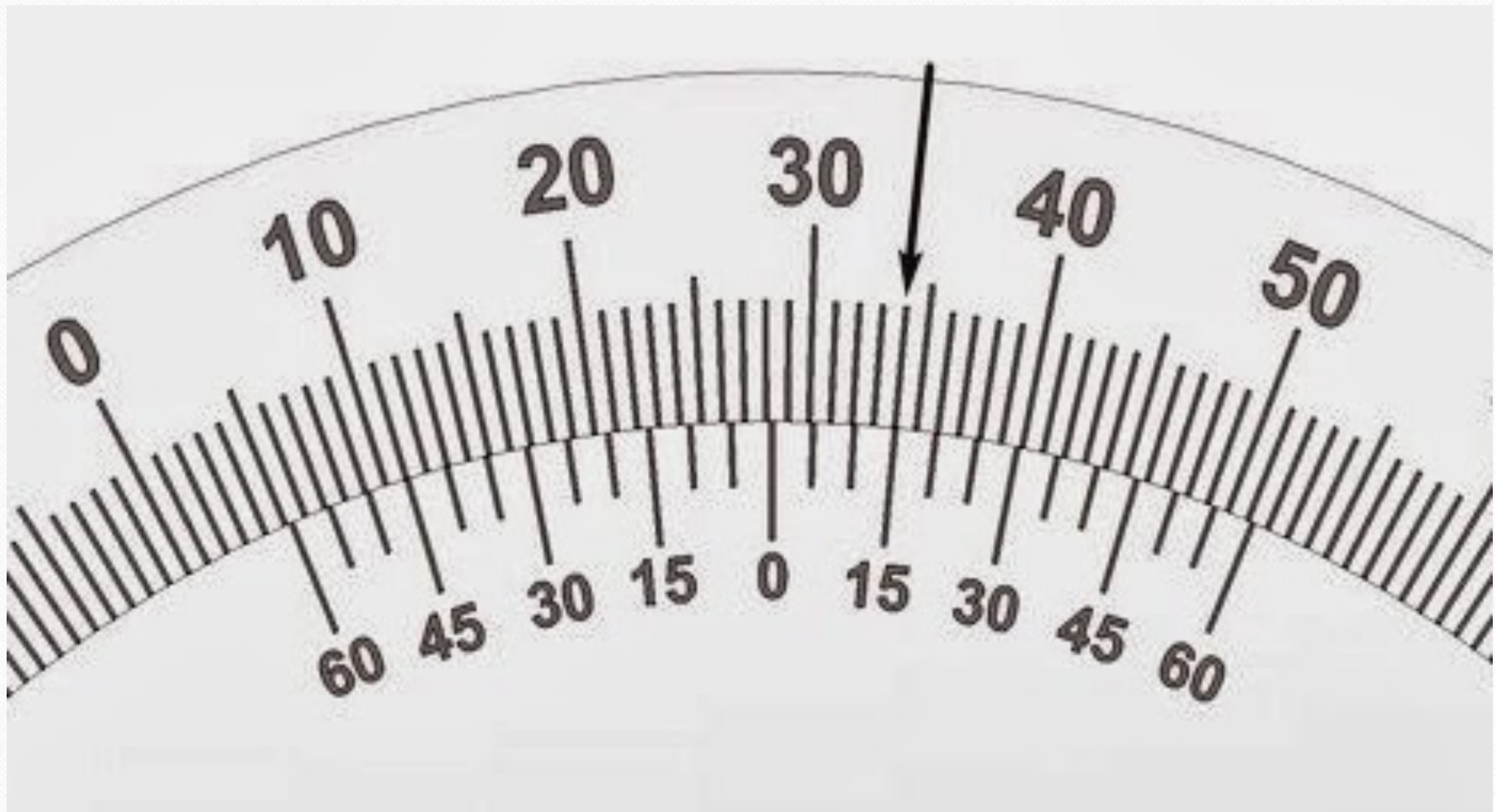
Sometimes, confusion arises regarding the direction in which the vernier has to be read.

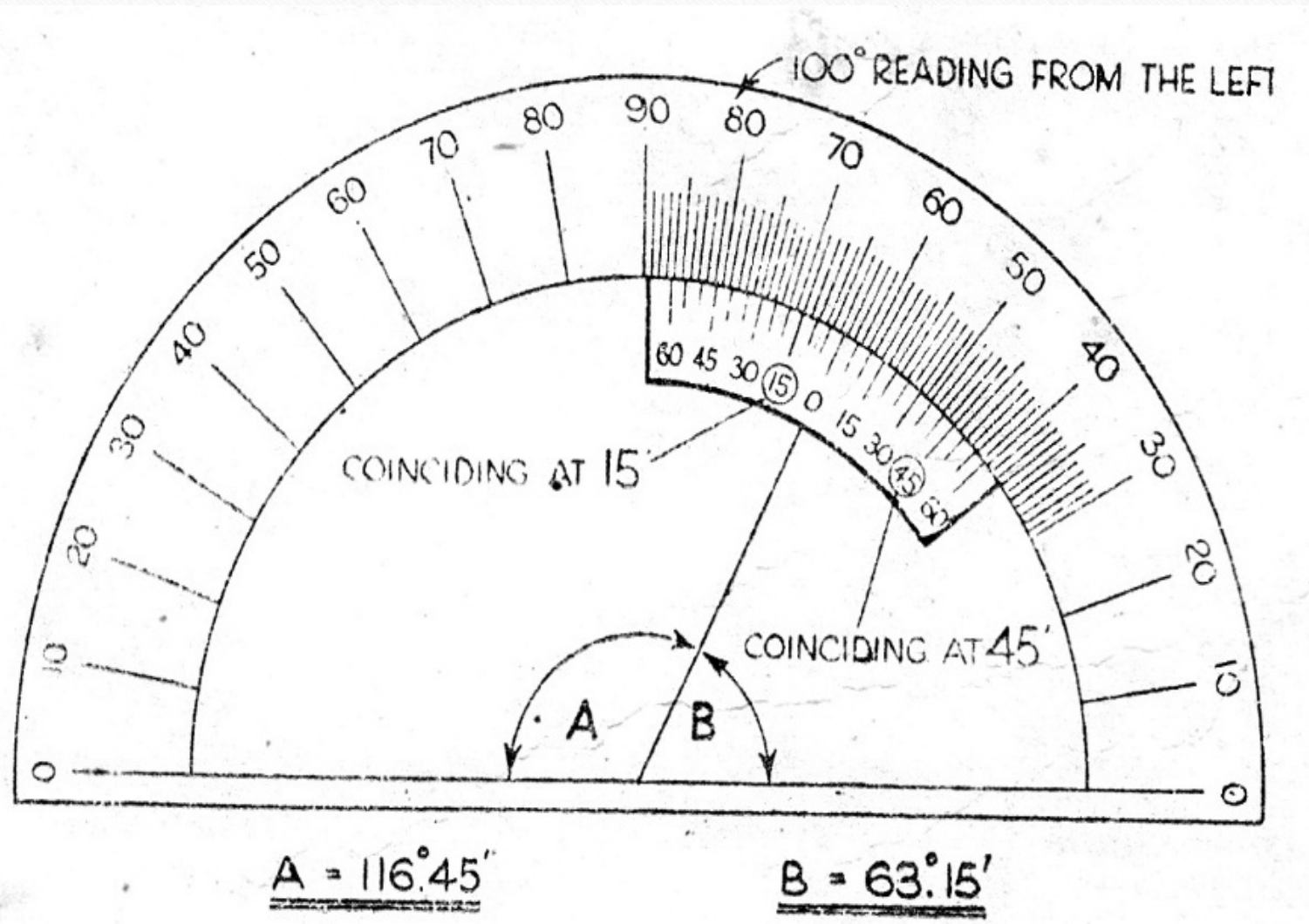
This confusion may crop up for the aforementioned example also. It is possible that a division on the right side of zero on the vernier scale may be coinciding with a division on the main scale (dial scale). In order to eliminate this confusion, we follow a simple rule. *Always read the vernier from zero in the same direction that you read the dial scale.* In the given example, the 10th division on the dial, which is close to the zeroth division on the vernier, is to the left of the zeroth division on the dial scale.

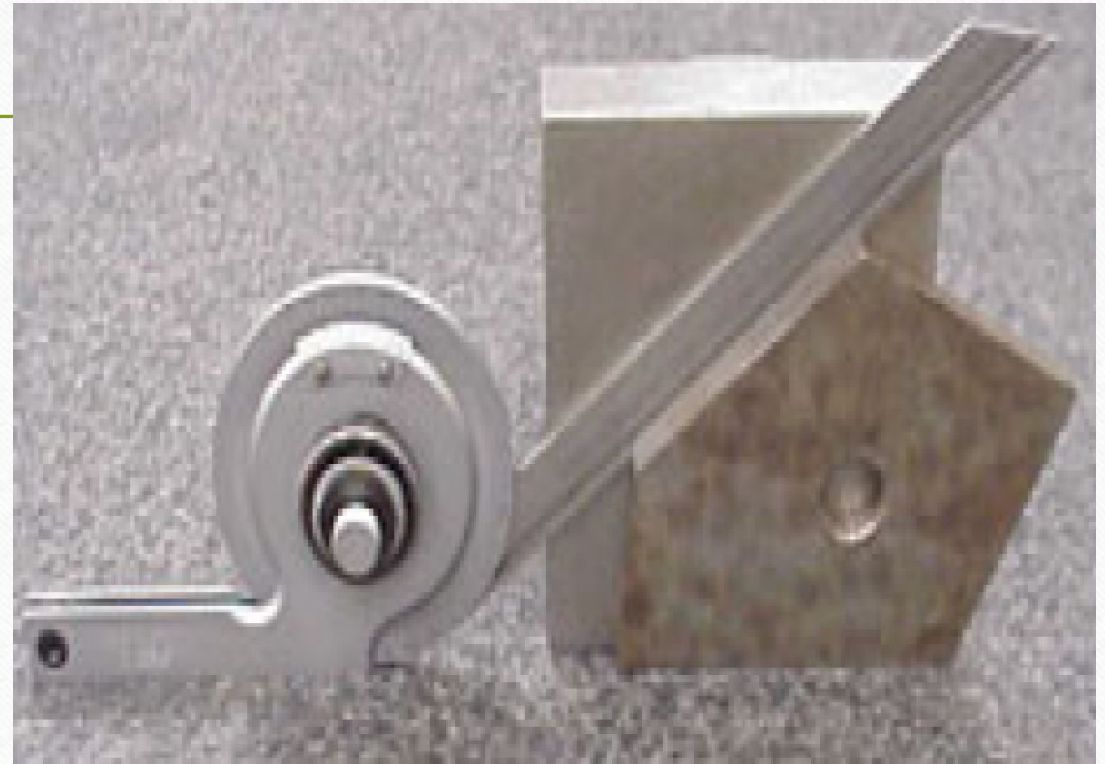
Main scale and vernier scale





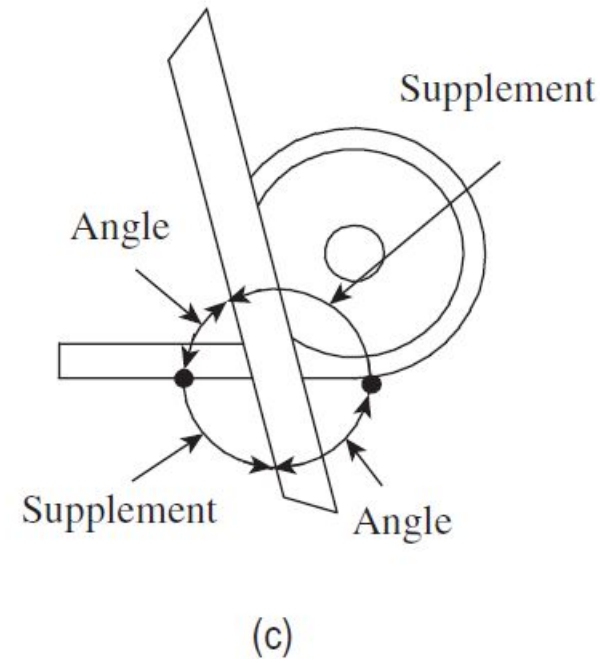
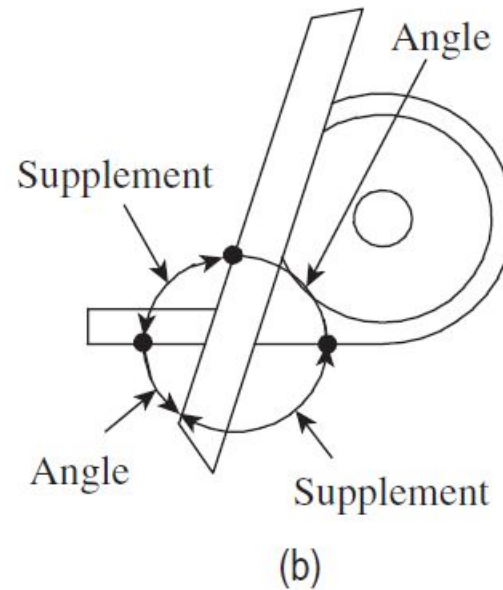
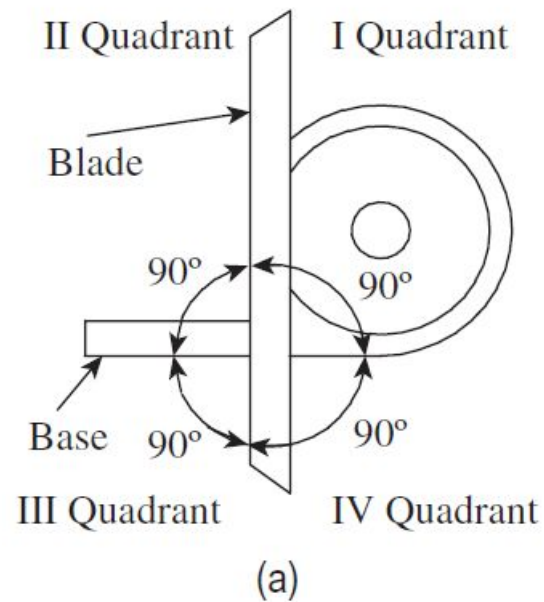






Angles and their Supplements

- Since a universal bevel protractor can measure both acute and obtuse angles, care should be exercised to clearly differentiate between the angle being indicated on the scale and its supplement. The dial gauge is graduated from 0° to 90° in four quadrants, as shown Fig (a). Figure (a) illustrates the orientation of the blade with respect to the base when the protractor is set to 90° . The zeroth division on the vernier coincides with the 90° division on the dial scale.



- Suppose the blade is turned clockwise, as in Fig.(b), the angles read directly are those that are formed from the blade to the base counterclockwise. Thus, if the angles of a work part are being measured in quadrant I or III, the angles can be read directly from the scale. On the other hand, if the angles of a work part are being measured in quadrant II or IV, the actual angles are given by their supplements. In other words, the value of the angle is obtained by subtracting the angle indicated by the scale from 180° . Both these angles, obviously, are obtuse angles. Further, (c) illustrates a situation when the blade is turned counterclockwise. Here also, angles can be directly read only in two quadrants, namely the second and the fourth. These angles are formed in the clockwise direction from the blade to the base, and are acute angles. The supplements in I and III quadrants give the obtuse angles of the work parts, which are held in these quadrants.

FACE STANDARD ANGULAR MEASURING DEVICES

- Face standard angular measuring devices include angle gauges and sine bars.
- The measurements are done with respect to two faces of the measuring instruments.
- Precision obtained in such instruments is more than the precision obtained in line standard angular measuring devices.

Sine bar

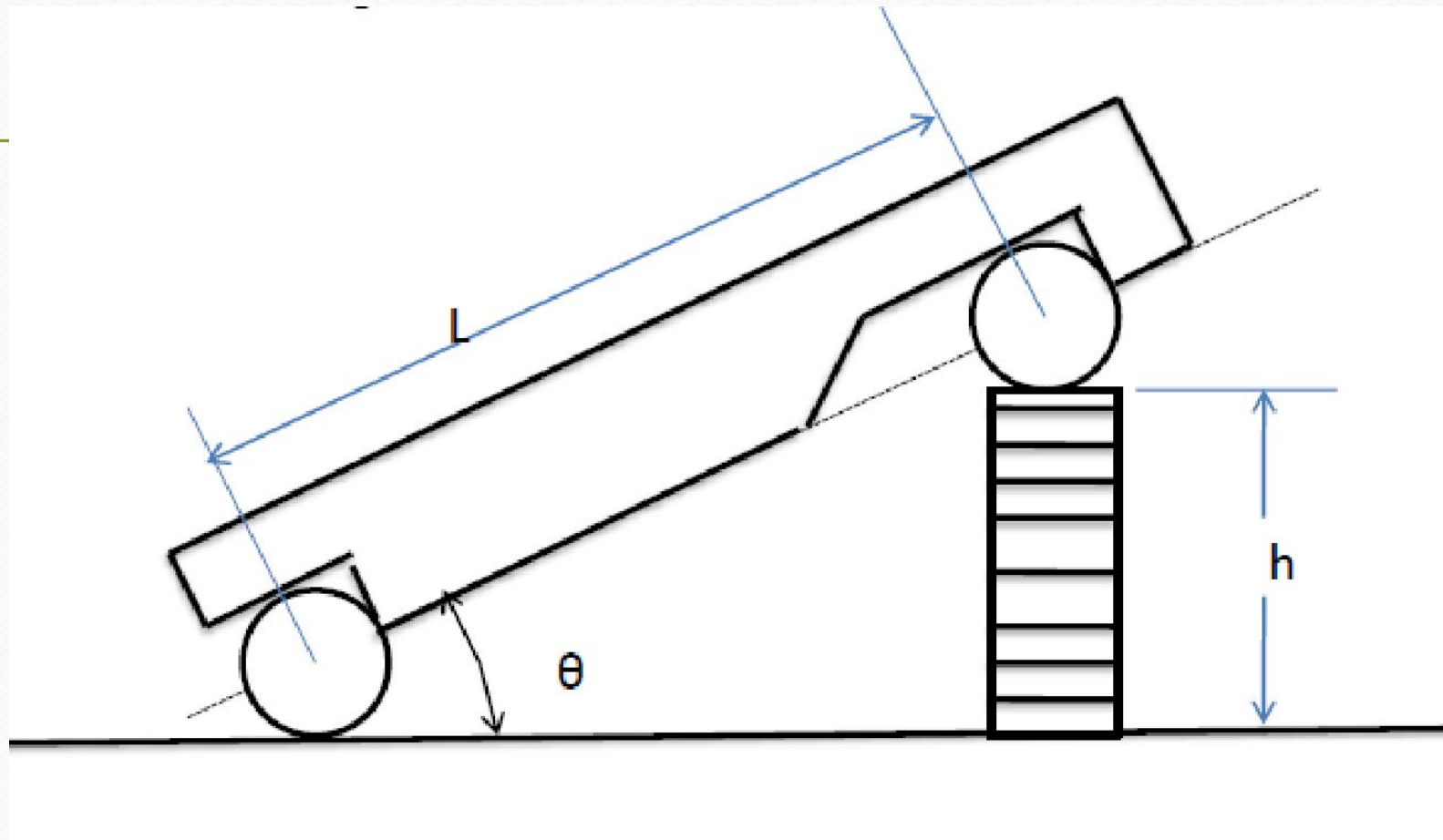
- It consists of a steel bar and two rollers.
- It is made of high carbon steel, high chromium corrosion resistant steel suitably hardened, precision ground and stabilized.
- Two rollers are attached to the bar at each end.
- The axes of these rollers are parallel to each other and also to the upper surface of the bar.
- The normal distance between the axes of the rollers is exactly 100mm, 200mm or 300mm.

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- Sine bars are graded as A grade and B grade.
 - A grade sine bar are made with an accuracy of 0.001 mm/m of length.
 - B grade sine bar with an accuracy of 0.002 mm/m of length.

Principle of sine bar

- The principle of operation of a sine bar is based on the laws of trigonometry.
- To set a given angle, one roller of the bar is placed on the surface plate and the combination of slip gauges is inserted under the second roller.
- If 'h' is the height of the combination of slip gauges and 'L' the distance between the roller centres,
- $\sin \theta = h/L$ or $\theta = \sin^{-1} (h/L)$
- The angle is measured by indirect method as a function of sine; the device is called sine bar.

Principle of sine bar

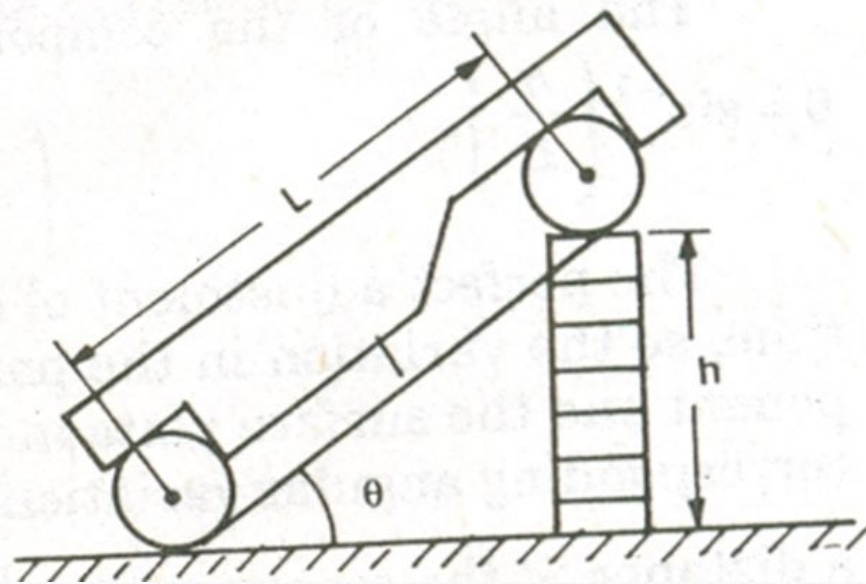


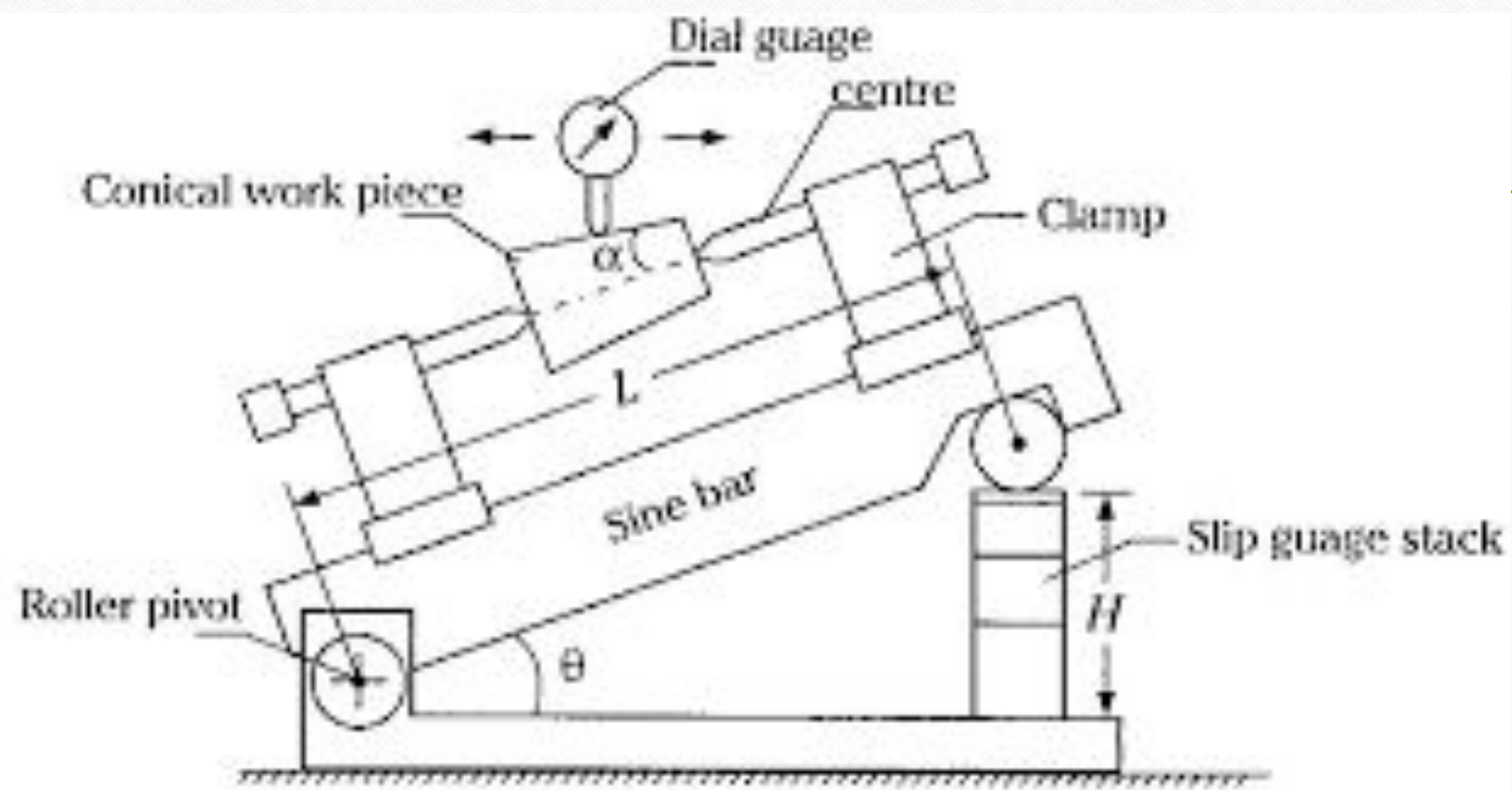
Use of sine bar

1. Locating any work to a given angle

- To set the given angle, the surface plate is assumed to be perfectly flat, so that the surface can be treated as horizontal.
- One roller of the sine bar is placed on the surface plate and a combination of slip gauges is inserted under the second roller.
- Let, h be the height of slip gauge combination and the sine is to be set at an angle θ .

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- Then $\sin \theta = h/l$,
 - where l is the distance between the centre of the rollers.
 - Thus knowing θ , h can be found out and any work could be set at this angle, as the top face of the sine bar is inclined at angle θ to the surface plate.



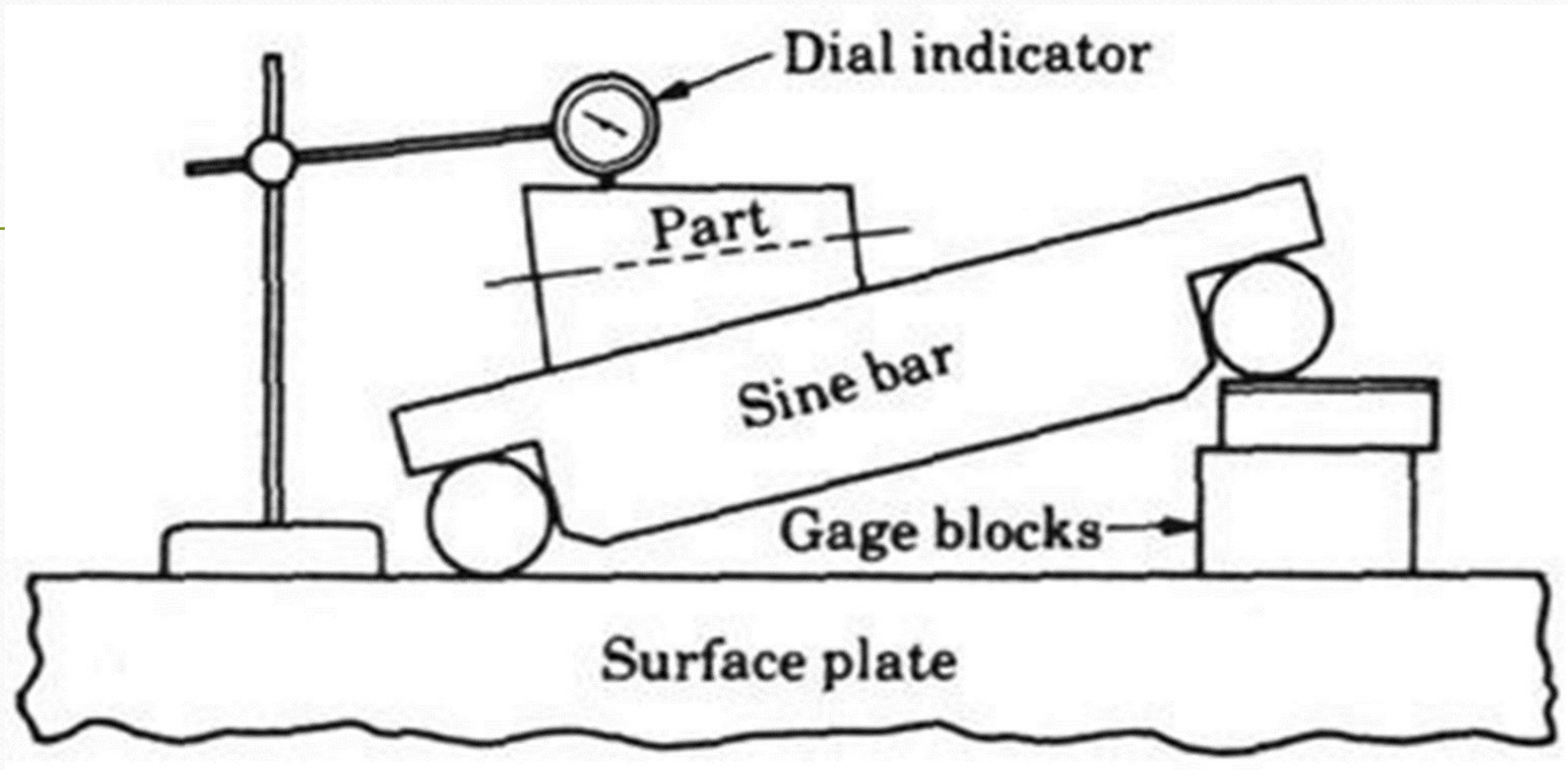


2. Checking or measuring unknown angles

- When component is of small size, for measuring unknown angle it is necessary to first find the angle approximately with the help of a bevel protractor.
- The sine bar is then set up at that nominal (approximate) angle on a surface plate by suitable combination of slip gauges.

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- The component to be checked is placed over the surface of the sine bar (If necessary the component may be clamped with the angle plate).
 - The dial gauge is then set at one end of the work and moved along the upper surface of the component.
 - If there is a variation in parallelism of the upper surface of the component and the surface plate, it is indicated by the dial gauge.

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- The combination of the slip gauges is so adjusted that the upper surface of the component is truly parallel with the surface plate.
 - The angle of the component then calculated by the relation $\theta = \sin^{-1} (h/L)$.
 - The perfect adjustment of slip gauge combination requires too much time, so the variation in the parallelism of the upper surface of the component and the surface plate indicated by the dial gauge is converted into corresponding angular variation.

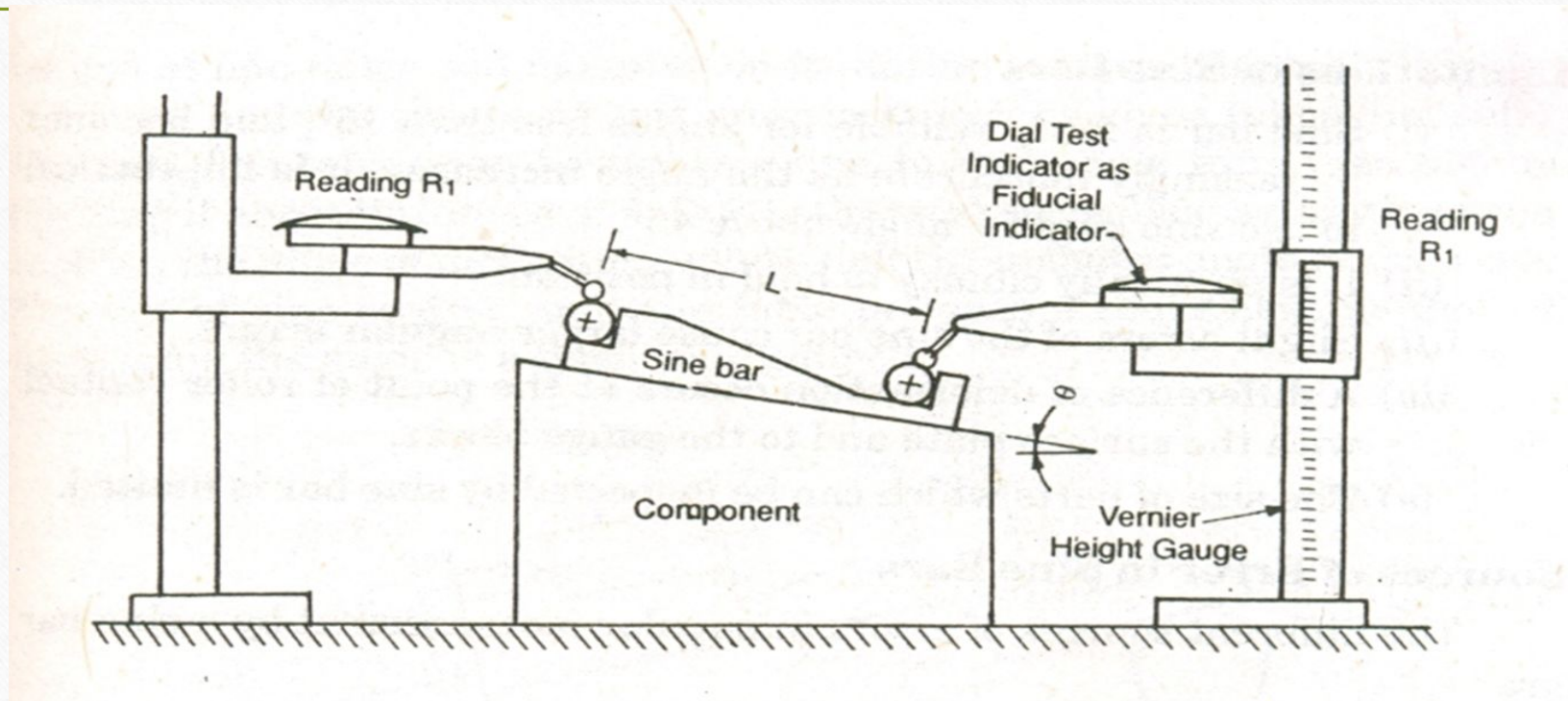


3. When the component is of large size / heavy.

- In such cases, the component is placed over a surface plate.
- The sine bar is placed over the component.
- The height over the rollers can then be measured by a vernier height gauge; using a dial test gauge mounted on the anvil of height gauge to ensure constant measuring pressure.

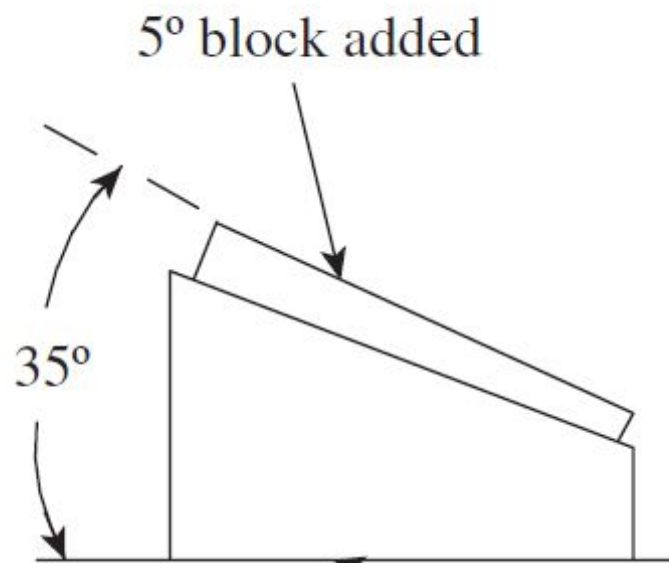
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- The anvil of height gauge is adjusted with probe of dial test gauge showing same reading for the top most position of rollers of sine bar.
 - The height gauge is thus used to obtain two readings for either of the rollers of sine bar.
 - If 'h' is the difference in the heights and ‘L’ distance between the roller centres of the sine bar, then $\theta = \sin^{-1} (h/L)$.

When the component is of large size / heavy

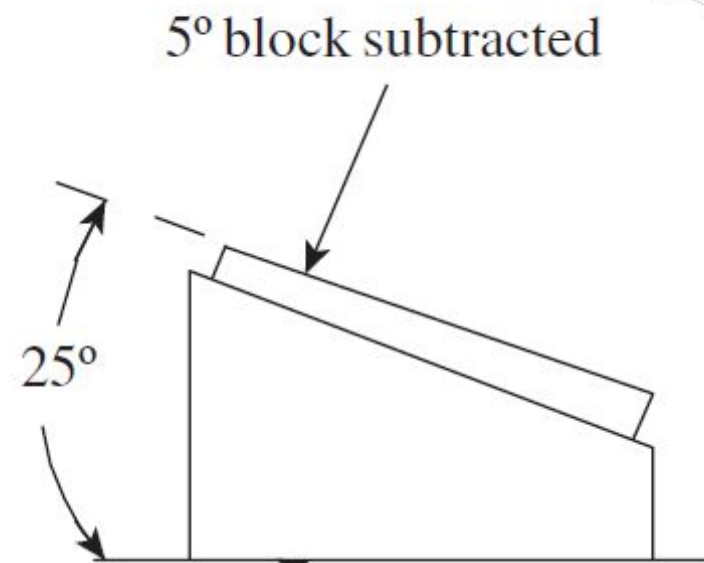


ANGLE GAUGES

Angle gauges, which are made of high-grade wear-resistant steel, work on a principle similar to slip gauges. While slip gauges can be built to give linear dimensions, angle gauges can be built to give the required angle. The gauges come in a standard set of angle blocks that can be wrung together in a suitable combination to build an angle. C.E. Johansson, who developed slip gauges, is also credited with the invention of angle gauge blocks. However, the first set of a combination of angle gauges was devised by Dr G.A. Tomlinson of the National Physical Laboratory, UK, in the year 1939, which provided the highest number of angle combinations. His set of 10 blocks can be used to set any angle between 0° and 180° in increments of $5'$. At the outset, it seems improbable that a set of 10 gauges is sufficient to build so many angles. However, angle blocks have a special feature that is impossible in slip gauges—the former can be subtracted as well as added.



(a)



(b)

Angle gauge block (a) Addition (b) Subtraction

Uses

Angle gauges are used for measurement and calibration purposes in tool rooms. It can be used for measuring the angle of a die insert or for inspecting compound angles of tools and dies. They are also used in machine shops for either setting up a machine (e.g., the revolving centre of a magnetic chuck) or for grinding notches on a cylindrical grinding machine.

MEASUREMENT OF INCLINES

- Inclination of a surface generally represents its deviation from the horizontal or vertical planes.
- Gravitational principle can be used in construction of measurements of such inclinations.
- Spirit levels and clinometer are the instruments of this category

Spirit Level

- Spirit level is one of the most commonly used instruments for inspecting the horizontal position of surfaces and for evaluating the direction and magnitude of minor deviation from that nominal condition.
- It essentially consists of a close glass tube of accurate form. It is called as the vial. It is filled almost entirely with a liquid, leaving a small space for the formation of an air or gas bubble.
- Generally, low viscosity fluids, such as ether, alcohol or benzol, are preferred for filling the vial..

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- The liquid due to its greater specific weight tends to fill the lower portion of the closed space.
 - Upper side of the vial is graduated in linear units.
 - Inclination of a surface can be known from the deviation of the bubble from its position when the spirit level is kept in a horizontal plane

Spirit levels are used for

- (i) measuring small angle or inclinations
- (ii) to determine the position of surface with respect to the horizontal position and
- (iii) to establish a horizontal datum.

Spirit Level



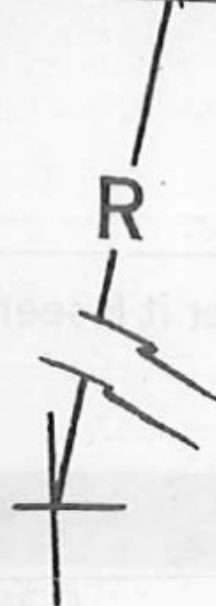
The spirit level consists of a sealed glass tube mounted on a base . The inside surface of the tube is ground to a convex barrel shape having large radius. The precision of the level depends on the accuracy of this radius of the tube. A scale is engraved on the top of the glass tube. The tube is nearly filled with either ether or alcohol, except a small air or vapour in the form of a bubble.

Principle

The bubble always tries to remain at the highest point of the tube. If the base of the spirit level is horizontal, the centre point is the highest point of the tube. So that, when the level is placed on a horizontal surface, the bubble rests at the centre of the scale. If the base of the level is tilted through a small angle, the bubble will move relative to the tube a distance along its radius corresponding to the angle.

SPIRIT LEVELS

Bent Type
Low Precision



Ground Type
High Precision



The longer the radius of curvature, the more precise the level will be.

Principle of spirit level

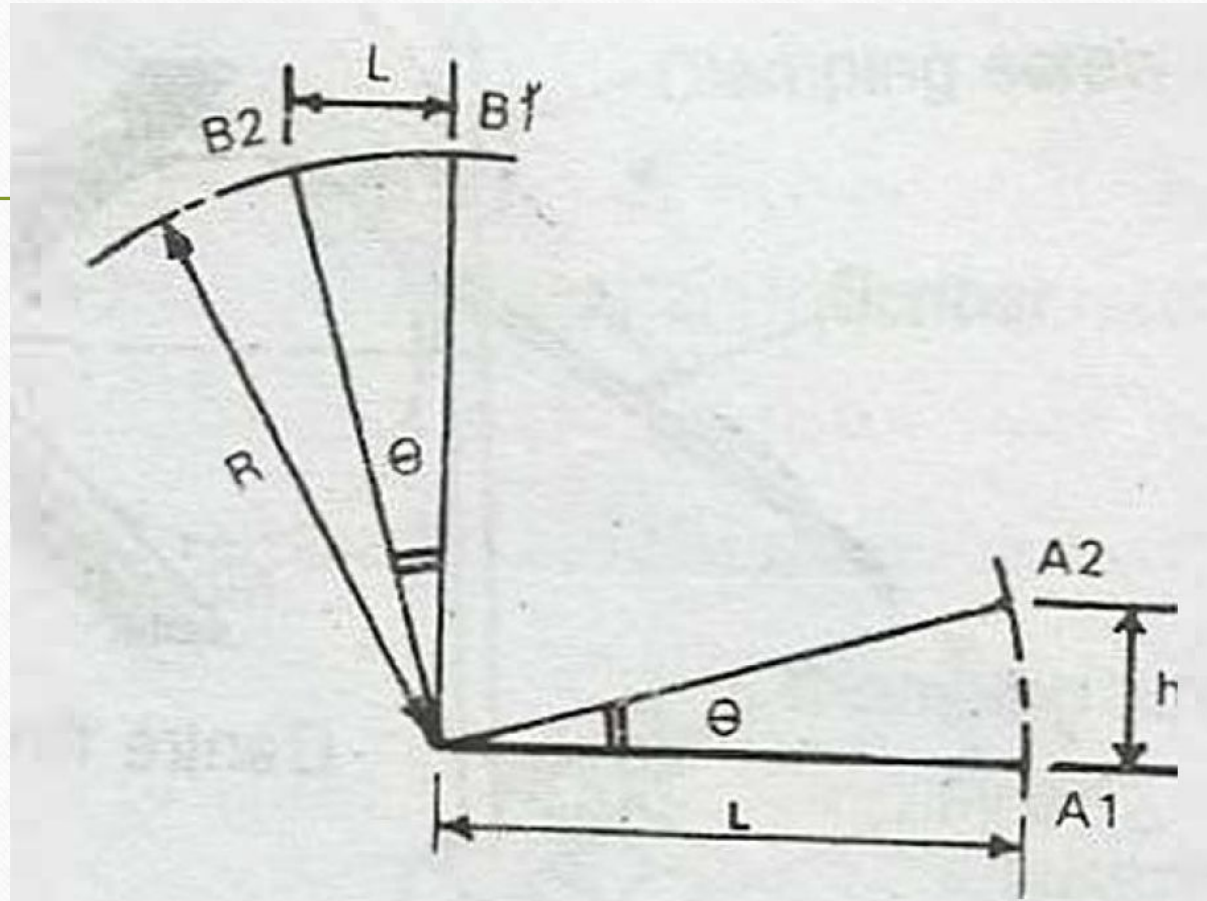


Fig shows two positions of the base of the level (OA1 and OA2) and corresponding positions of the bubble (B1, B2). When the base (OA 1) is horizontal, the bubble occupies position B1. Let ' θ ' be the small angle through which the base is tilted. The bubble now occupies the position B2. Let ' l ' be the distance travelled by the bubble along the tube and ' h ' the difference in heights between the ends of the base, then,

$$l = R \theta \quad \text{and} \quad h = L \theta$$

$$\text{Therefore, } \theta = \frac{l}{R} = \frac{h}{L}$$

$$\text{or} \quad l = h \cdot \frac{R}{L}$$

where, R = radius of curvature of the tube
 L = length of base

Sensitivity of spirit level

Sensitivity of spirit level is the angle of tilt in seconds that will cause the bubble to move through one division.

- (i) ~~A 10 second level means that tilting the level through an angle of 10 seconds to the horizontal will move the bubble by one deviation.~~
- (ii) A level with a sensitivity of 0.05 mm/m means that if the level were placed one metre long straight edge, and that if one end of that straight edge was raised by 0.05 mm, then the bubble will move one division.

It depends upon the radius of curvature, length of bubble and internal radius of the tube.

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Uses

Spirit levels are used

- (i) to measure small angles or inclinations and,
- (ii) to test straightness and flatness of surfaces.

When using a spirit level to determine horizontal surface it should be used in two directions at right angles to check the horizontal plane.

Clinometer

- A clinometer is a special case of application of spirit level for measuring, in the vertical plane, the incline of a surface in relation to the basic horizontal plane, over an extended range.
- The main functional element of a clinometer is the sensitive vial mounted on a rotatable disc, which carries a graduated ring with its horizontal axis supported in the housing of the instrument.
- The bubble of the vial is in its centre position, when the clinometer is placed on a horizontal surface and the scale of the rotatable disc is at zero position.

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- If the clinometer is placed on an incline surface, the bubble deviates from the centre. It can be brought to the centre by rotating the disc.
 - The rotation of the disc can be read on the scale. It represents the deviation of the surface over which the clinometer is placed from the horizontal plane.

While the spirit level is restricted to relatively small angles, clinometers can be used for much larger angles.

It comprises a level mounted on a frame so that the frame may be turned to any desired angle with respect to a horizontal reference.

Clinometers are used to determine straightness and flatness of surfaces.

They are also used for setting inclinable tables on jig boring machines and angular jobs on surface grinding machines.

They provide superior accuracy compared to ordinary spirit levels.

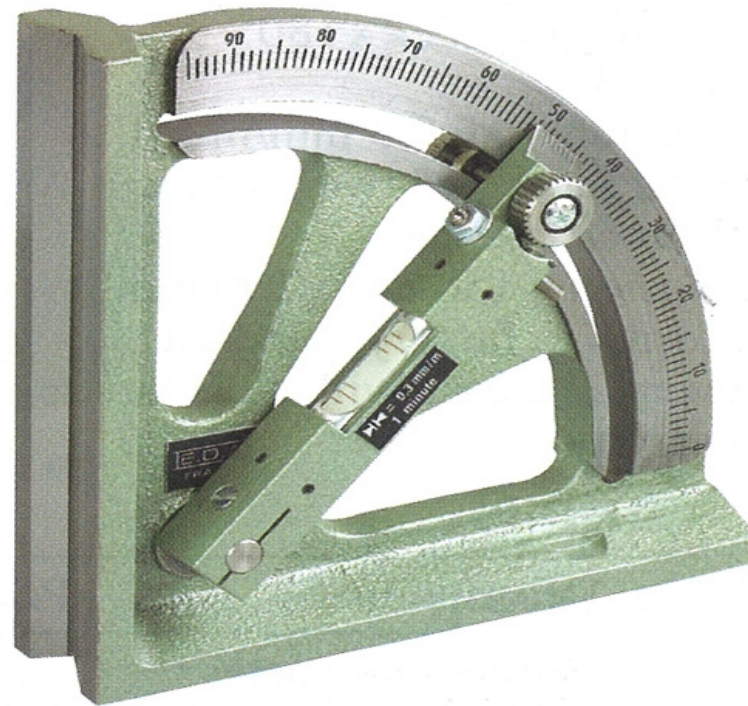
Vernier Clinometer

It consists of a spirit level mounted on a rotary member earned in a housing. One face of the housing forms the base of the instrument. There is a circular scale on the housing.

The angle of inclination of the rotary member relative to the base can be measured by a circularv. scale. The scale may cover the whole circle or only part of it.

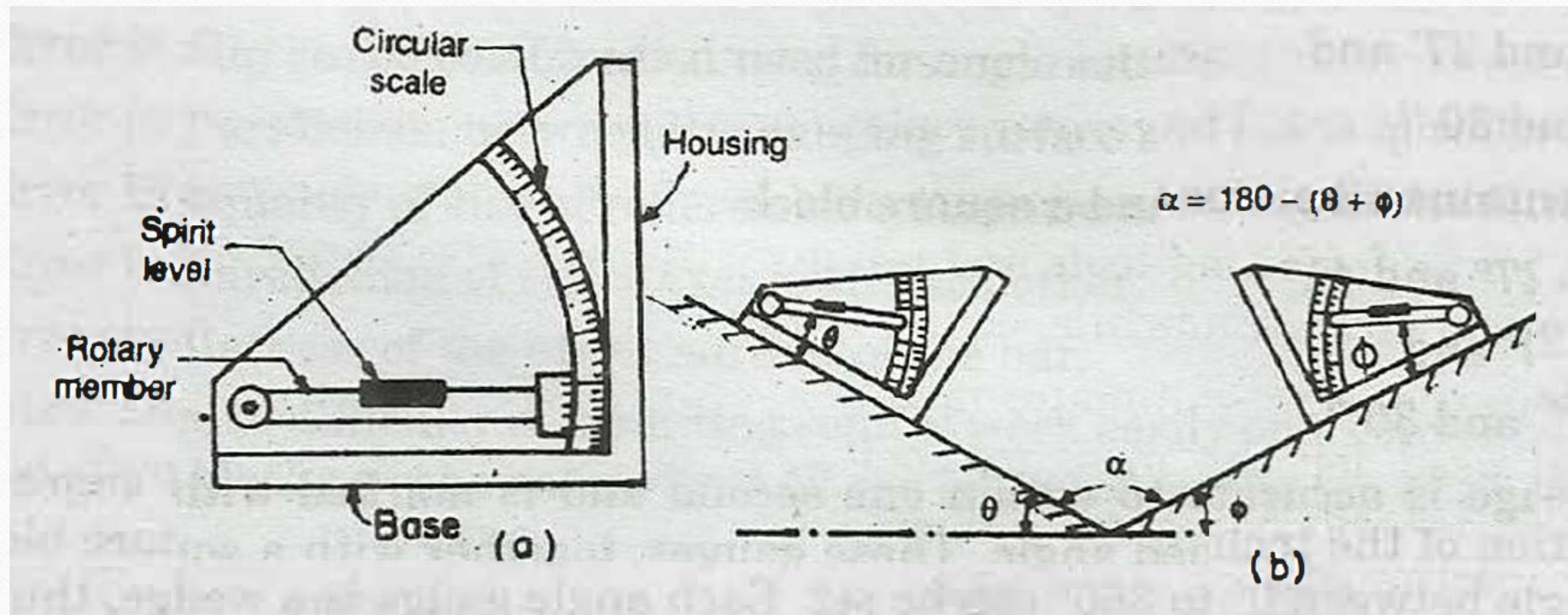
Clinometers are generally used to determine the angle included between two adjacent faces of a work-piece. The base of the instrument is placed on one of the surfaces and rotary member is adjusted till zero reading of the bubble is obtained. The angle of rotation is then noted on the circular scale against an index. The instrument is then placed on the other surface and the reading is taken in the similar manner.

Clinometer



If θ and φ are the reading of the instrument, the included angle between the surfaces,

$$\alpha = 180 - (\theta - \varphi)$$



Angle gauge

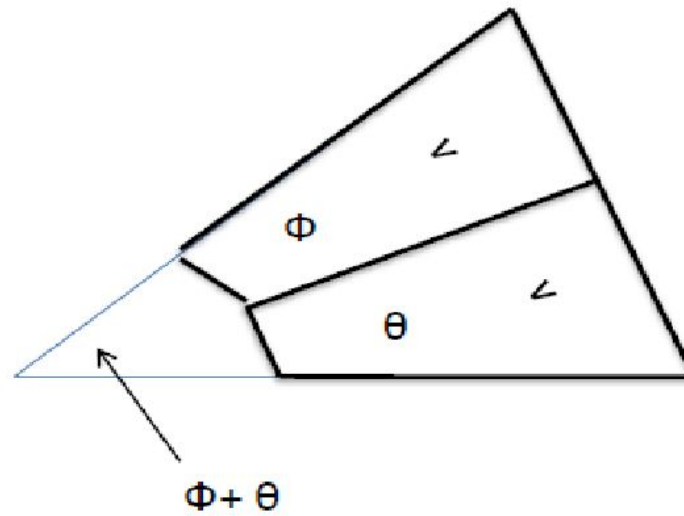
- Angle gauges were developed by Dr. Tomlinson in 1941, which enable any angle to be set to the nearest 3”.
- These are pieces of hardened steel.
- The measuring faces are lapped and polished to a high degree of accuracy and flatness.
- They are 75mm long and 16mm wide and are available in two sets.

□ One set consists of 12 pieces and a square block, in three series of values of angle viz

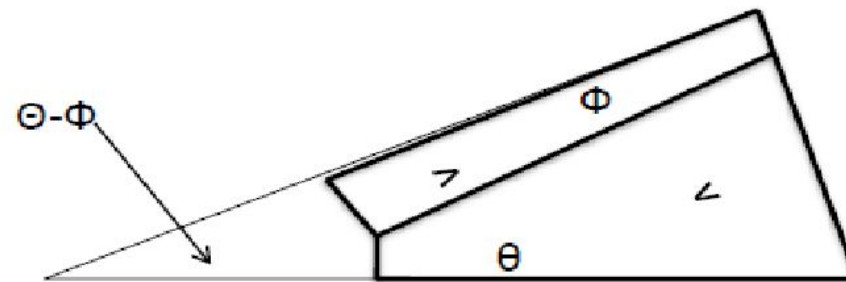
- $1^\circ, 3^\circ, 9^\circ, 27^\circ$ and 41°
- $1', 3', 9'$ and $27'$ and
- $6'', 18''$ and $30''$.

□ Another set contains 13 pieces and a square block

- $1^\circ, 3^\circ, 9^\circ, 27^\circ$ and 41°
- $1', 3', 9'$ and $27'$ and
- $3'', 6'', 18''$ and $30''$.



Addition and subtraction of angle gauges



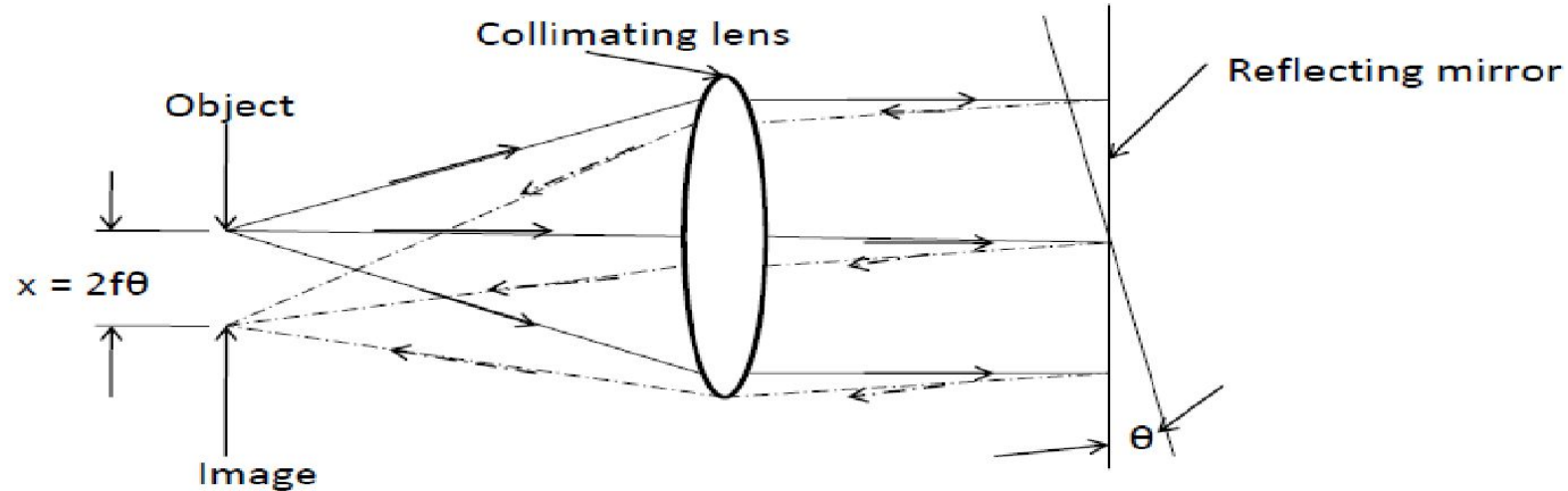
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- Each angle gauge is marked with engraved V which indicates the direction of included angle.
 - When the angles of individual angle gauges are to be added up then the V of all should be in line and when any angle is to be subtracted, its engraved V should be in other direction

Auto-collimator

- Auto-collimator is an optical instrument used for the measurement of small angular differences, changes or deflection, plane surface inspection etc.



Principle of auto-collimator



Where f = focal length of lens

θ = angle of inclination of reflecting mirror

If a light source is placed in the focus of a collimating lens, it is projected as a parallel beam of light. If this beam is made to strike a plane reflector, kept normal to the optical axis, it is reflected back along its own path and is brought to the same focus. If the reflector is tilted through a small angle ' θ ', the parallel beam is deflected twice that angle and is brought to a focus in the same plane as the light source, but to one side at a distance $x = 2f\theta$

where f = focal length of lens

θ = angle of inclination of reflecting mirror.

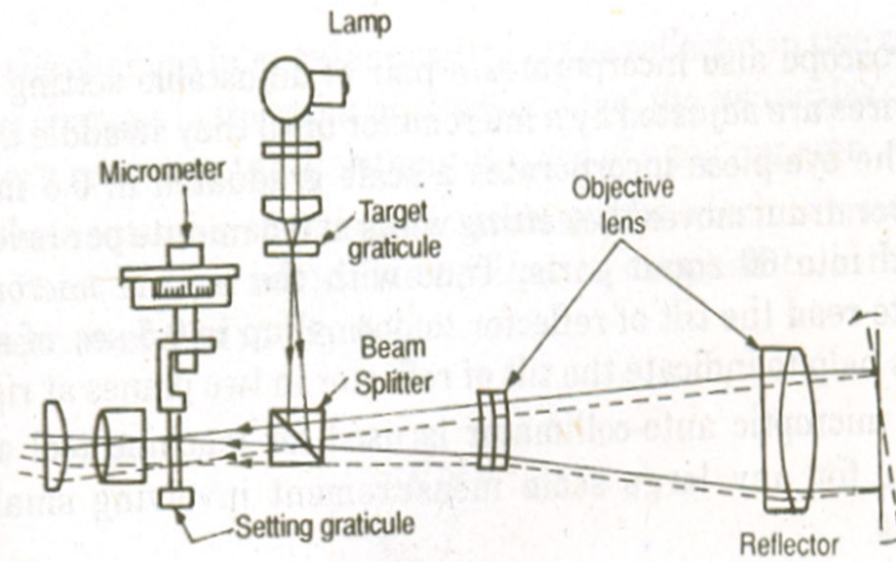
construction

- In auto-collimator there are three parts

- Micrometer microscope

- Lighting unit

- Collimating lens



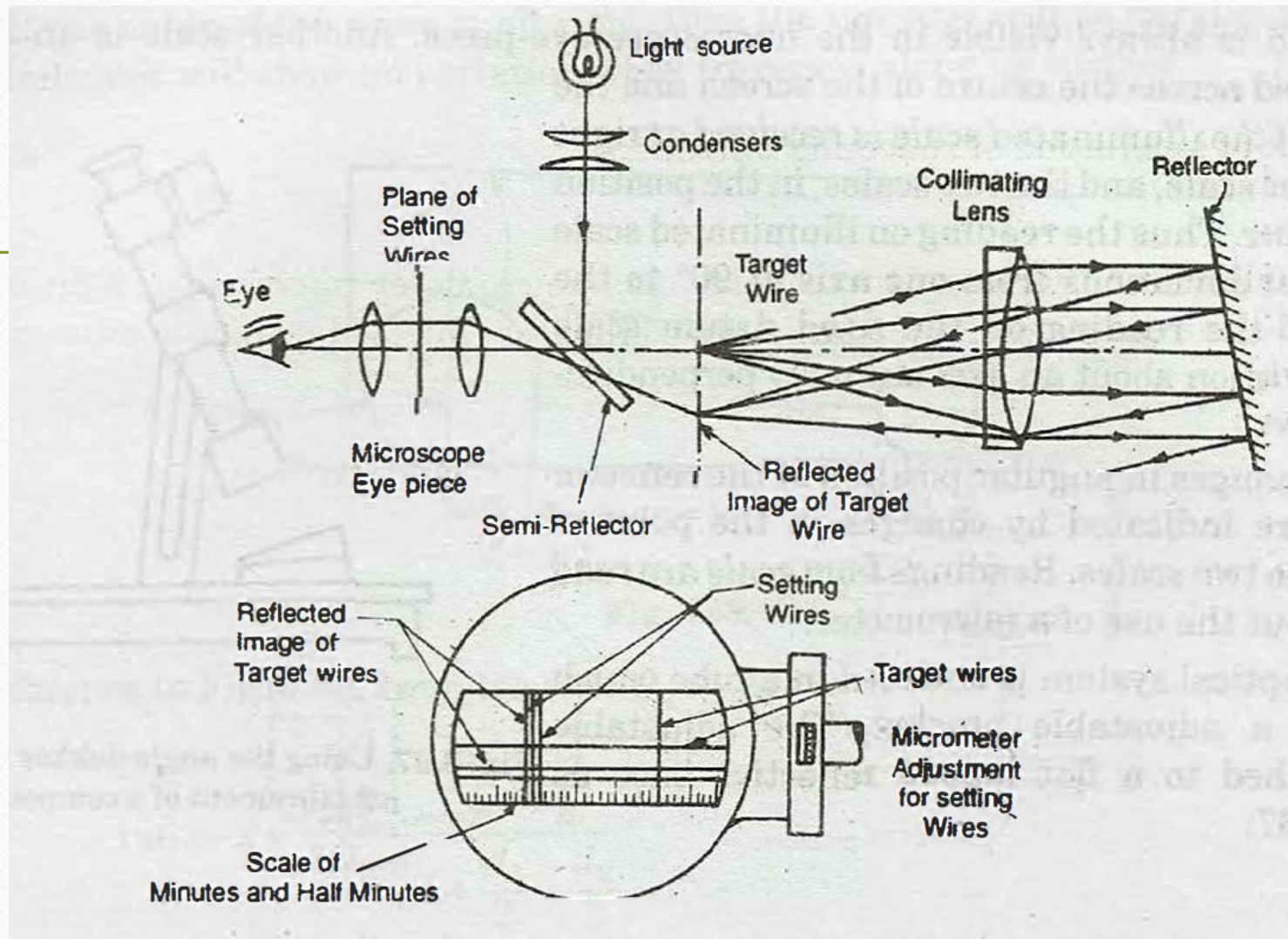
- ← Optical return path reflector square
- ← Optical return path reflector tilted

Line diagram of an injected graticule autocollimator

Fig. 6.20. Construction of auto-clinometer

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- A 45 deg transparent beam splitter reflects the light from graticule towards the objective (collimating lenses).
 - The image seen after reflection in the external reflector, whose angular variations are being measured is formed by the light from the objective lens.
 - This light passes through the beam splitter and the image is picked up by the microscope.

Microptic Auto-Collimator



In this instrument a pair of target wires in the focal plane of the collimating lens is illuminated by a lamp and condenser at the site and their image projected.

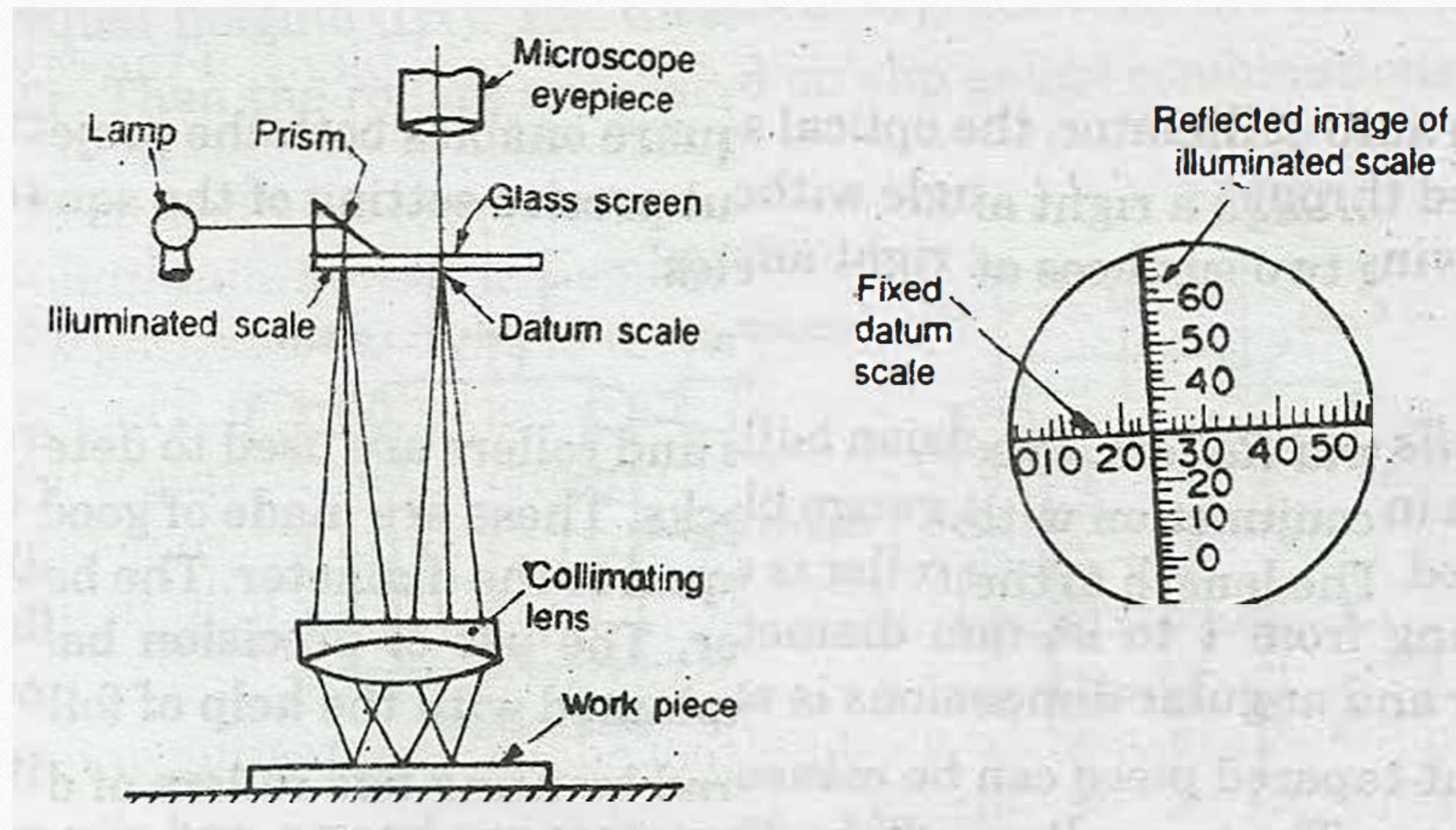
The projected image strikes a plane reflector and the reflection of the image is brought to a focus in the plane of target wires. The target wires and their image are viewed simultaneously through a microscope eye-piece. The microscope also incorporates a pair of adjustable setting wires. The setting wires are adjusted by a micrometer until they straddle the reflected image. The eye-piece incorporates a scale graduated in 0.5 minute. The micrometer drum moves the setting wires at 0.5 minute per revolution and is divided into 60 equal parts. Thus with the help of micrometer it is possible to read the tilt of reflector to normal up to 0.5 sec. of an arc. The two wires help to indicate the tilt of reflector in two planes at right angles.

Applications of auto-collimator

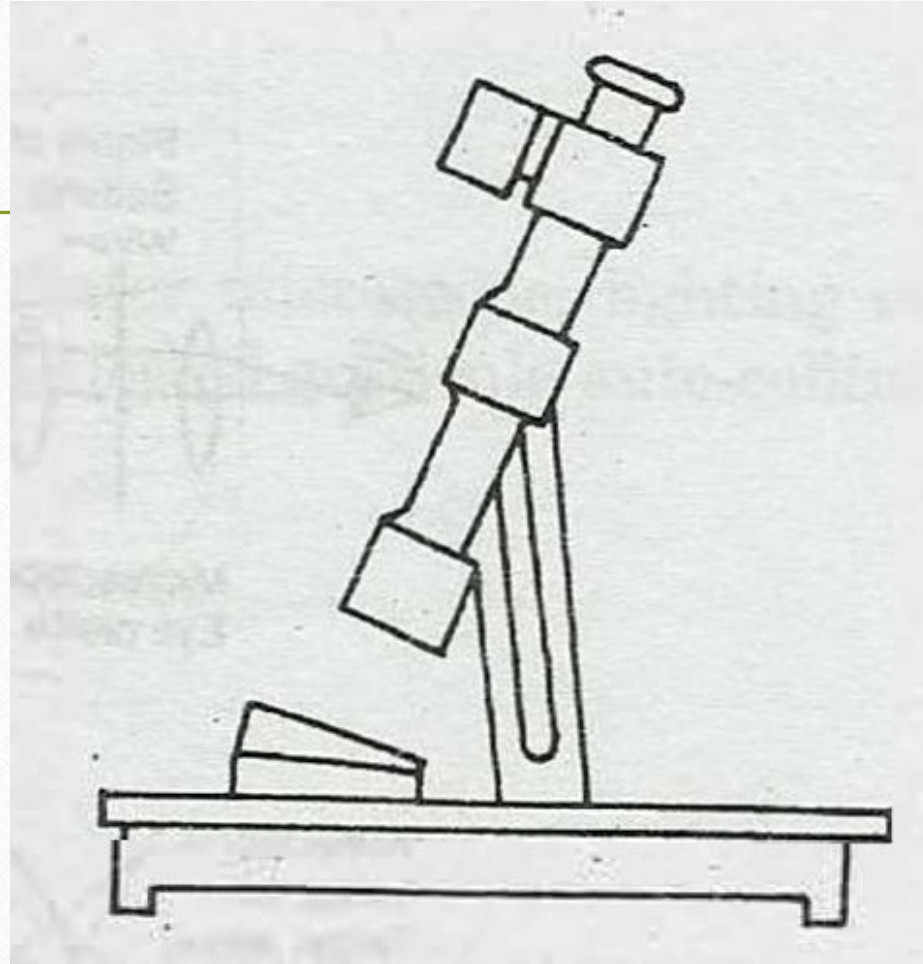
- the measurement of straightness and flatness of surfaces
- precise angular indexing in conjunction with polygons
- comparative measurement using master angles
- assessment of squareness and parallelism of components
- the measurement of small linear dimensions, and
- for machine tool adjustment setting etc.

Angle Dekkor

This is a type of auto-collimator. It consists of microscope, objective (collimating) lens and two scales engraved on a glass screen which is placed in the focal plane of the objective lens. One of the scales, called datum scale; is horizontal and fixed. It is engraved across the centre of the screen and is always visible in the microscope eye piece. Another scale is an illuminated vertical scale fixed across the centre of the screen and the reflected image of the illuminated scale is received at right angles to this fixed scale, and the two scales, in the position intersect each other. Thus the reading on illuminated scale measures angular deviations from one axis at 90° to the optical axis, and the reading on the fixed datum scale measures the deviation about an axis mutually perpendicular to the other two.



Using the angle dekkor for the angle measurement of a component



Taper Measurement
